AI WORKSHOP · KOOD/VÕRU · 26-10-2026

AI under the hood

How it works, and how to run it on your own laptop

PLAN

The next 70 minutes

1

Under the hood

Tokens, prediction, training

~20 min

2

Run it locally

Ollama, models, hardware

~10 min

3

Demos

Terminal, API, a tiny agent

~20 min

4

Your turn

Try it yourself

~15 min

UNDER THE HOOD

Where LLMs fit

Artificial intelligence
Computers doing things that look smart

Machine learning
Learn patterns from data instead of hand-written rules

Deep learning
Machine learning with large neural networks

Large language models
Neural networks trained on huge amounts of text

UNDER THE HOOD

An LLM predicts the next token

The capital of Estonia is ___



Illustrative numbers

UNDER THE HOOD

Text becomes tokens, tokens become numbers




```
[ 42, 14, 53, 12437, 2739, 374, 264, 11058, 2978 ]
```

- Rule of thumb: 1 token ≈ ¾ of an English word
- Estonian often needs more tokens per word
- Why **“how many r's in strawberry?”** is hard

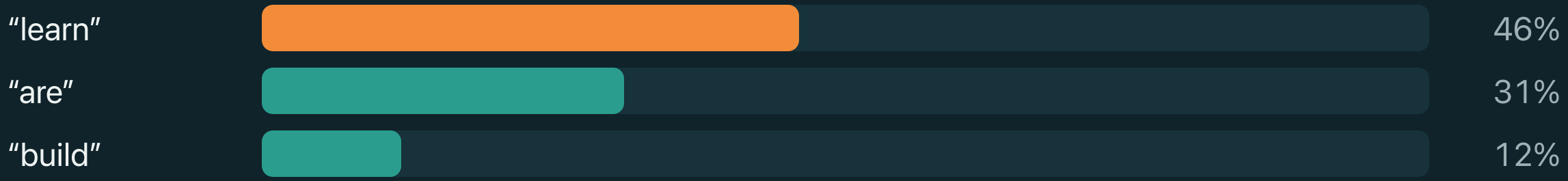
Illustrative split and IDs. Try a real tokenizer: tiktokenizer.vercel.app

UNDER THE HOOD

Generating text is a loop

Read the text so far → Score every possible next token → Pick one → Add it and repeat

Kood/Võru students



Next token | Reset | Illustrative probabilities

UNDER THE HOOD

Temperature: how adventurous the pick is

The best thing about Võru is the ___



Temperature: **1.0**

0 = always the top token

Sample 10 times

Toy scores, real maths (softmax with temperature)

UNDER THE HOOD

Where the knowledge comes from

1

Pre-training

Read huge amounts of text. Guess the next token. Adjust. Repeat.

2

Fine-tuning

Learn from examples of questions and good answers.

3

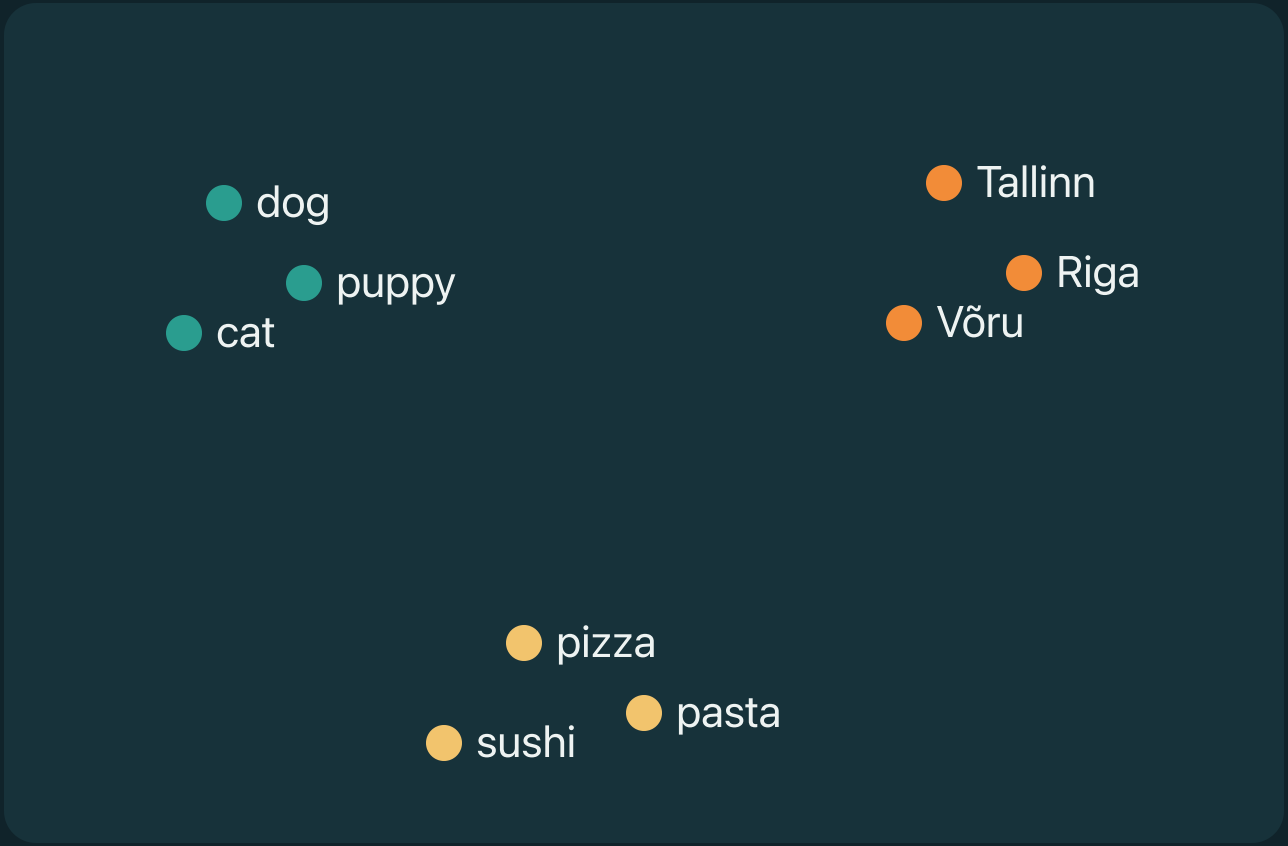
Feedback

Rated answers teach it which replies people prefer.

8B = 8 billion parameters: the numbers training adjusts. More parameters usually means smarter, but bigger and slower.

UNDER THE HOOD

Meaning becomes position



- Each token becomes a list of numbers
- Similar meaning, nearby numbers
- Used for search over your own docs

Illustrative 2D view. Real vectors have hundreds of dimensions.

UNDER THE HOOD

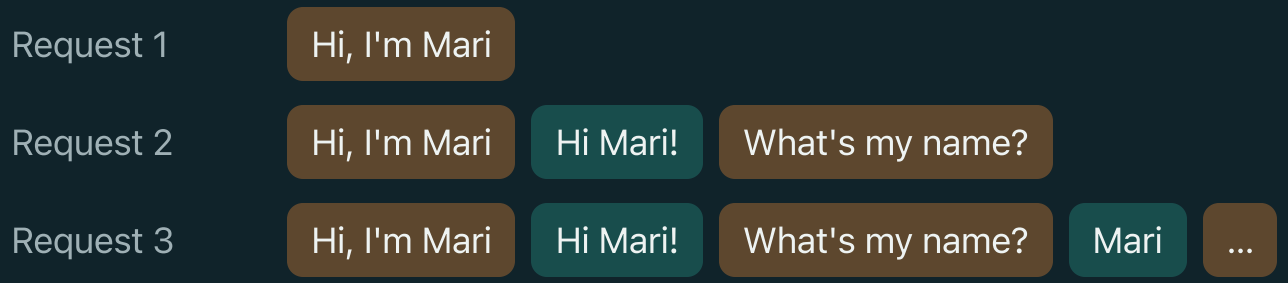
Attention: which words matter for this one?

The animal didn't cross the street because it was too tired.

- Every token looks at every other token
- “it” pays most attention to “animal”
- This is the “T” in GPT: **Transformer**

UNDER THE HOOD

The model has no memory



- Every request sends the whole conversation again
- **Context window**: the most tokens one request can hold

UNDER THE HOOD

What it's bad at

Making things up

Sounds confident when wrong. Plausible is not the same as true.

Counting and exact maths

It sees tokens, not letters or digits.

Recent events

Knowledge stops at its training cutoff.

Knowing what it doesn't know

Check its work. Give it context. Ask for sources.

PART 2

Run it on your own laptop

No subscription needed

RUN IT LOCALLY

Why run a model locally?

1

Free

No subscription, no API bill

2

Private

Your data stays on your machine

3

Offline

Works without internet

4

Hackable

A local API you can build on

Trade-off: smaller models, less smart than the big cloud ones. Needs RAM.

RUN IT LOCALLY

Install Ollama

macOS / Windows

Download from **ollama.com/download**

Linux

```
curl -fsSL
https://ollama.com/install.sh | sh
```

Then

```
# download a small model
ollama pull llama3.2:3b

# chat with it
ollama run llama3.2:3b
```

Slow Wi-Fi or 8 GB RAM? Use `llama3.2:1b` or `gemma3:1b`.

RUN IT LOCALLY

Which model fits my laptop?

YOUR RAM	TRY	DOWNLOAD (APPROX.)
8 GB	`llama3.2:1b` · `gemma3:1b` · `llama3.2:3b`	0.8 – 2 GB
16 GB	`llama3.1:8b` · `qwen2.5:7b`	~5 GB
32 GB+	`qwen2.5:14b`	~9 GB

Quantization: weights stored in fewer bits. Much smaller, slightly less accurate.

More models: ollama.com/library

RUN IT LOCALLY

Cheat sheet

```
ollama list          # models you have
ollama ps            # models loaded now
ollama show llama3.2:3b
ollama rm llama3.2:3b
```

```
# inside ollama run
/set parameter temperature 0
/set system "Answer in one sentence."
/clear
/bye
```

API on `http://localhost:11434`

PART 3

Demos

Switching to the terminal

DEMO 1 · TERMINAL

Chat with a local model

```
ollama run llama3.2:3b
```

Try

- What is today's date?
- Tell me three facts about Võru.
- How many r's are in “strawberry”?

Watch for: confident wrong answers

DEMO 2 · TERMINAL

Temperature for real

```
/set parameter temperature 0
Write a one-line slogan for Võru.

/set parameter temperature 1.5
Write a one-line slogan for Võru.
```

Same prompt, three times each. Compare.

DEMO 3 · TERMINAL

System prompt: set the rules

```
/set system "You are a pirate. Answer in one sentence."
What is a database?

/set system "You explain code to a 10-year-old."
What is recursion?
```

Hidden instructions sent before your message

DEMO 4 · BROWSER

Watch it generate, token by token

llama3.2:3b

Explain what a token is in two sentences.

Run

Stop

Talks to Ollama on localhost:11434

DEMO 5 · TERMINAL

It's just an API

```
curl http://localhost:11434/api/chat -d '{
  "model": "llama3.2:3b",
  "messages": [{"role": "user", "content": "Hi! Who are you?"}],
  "stream": false
}'
```

Script: `demos/api.sh`

DEMO 6 · TERMINAL

Memory is just a list

```
python3 demos/chat.py
python3 demos/chat.py --no-memory
```

- Tell it your name, then ask for it
- Without history, it forgets

~40 lines of Python, no libraries

DEMO 7 · TERMINAL

Embeddings: meaning as numbers

```
ollama pull nomic-embed-text
python3 demos/embeddings.py
```

- Turns sentences into vectors
- Scores how similar they are
- Similar meaning, high score, even with different words

DEMO 8 · TERMINAL

A tiny agent: the model asks, your code acts




```
python3 demos/tools.py "What is 1234 * 5678, and what day is it today?"
```

Coding agents are this loop with more tools

PART 4 · YOUR TURN

Try it yourself

1

Install

Ollama + `llama3.2:1b` or `3b`

2

Break it

Find a question it gets confidently wrong

3

Tune it

Same prompt at temperature 0 and 1.5

4

Shape it

System prompt: a study buddy for your current project

5

Build it

Add your own tool to `tools.py`

?

Stuck?

Pair up. Ask your neighbour.

WRAP UP

Remember

1

It predicts the next token

Trained on lots of text. Plausible, not always true.

2

Tokens, temperature, context

The knobs you control.

3

Local models are free

Good enough to learn and build with.

4

Agents are a loop

Model asks, code acts, repeat.

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Questions?

Thanks for having me